Prim's Algorithm for Minimum Spanning Tree (MST):

Definition:

Prim's Algorithm is a greedy algorithm used to find the Minimum Spanning Tree of a connected, weighted, undirected graph. It connects all vertices with minimum total edge weight without forming cycles.

Algorithm:

1. Start with any vertex.
2. Mark the starting vertex as selected.
3. Find the minimum-weight edge connecting a selected vertex to an unselected vertex.
4. Add the edge to the MST.
5. Mark the new vertex as selected.
6. Repeat steps 3–5 until all vertices are selected.
7. Calculate the total cost of the MST.

Out put:

```
Enter number of vertices: 4
Enter the adjacency matrix:
0 10 6 4
2 6 4 10
4 3 6 5
7 8 9 11

Edges in Minimum Spanning Tree:
1 - 4 : 4
1 - 3 : 6
3 - 2 : 3

Total cost of MST: 13
Execution time: 9.703636169433594e-05 seconds


...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

For an adjacency matrix implementation:

$$O(V^2)$$

Space Complexity

For an adjacency matrix implementation:

$$O(V^2)$$